2024- 2026

Mechanical Engineering Portfolio

Will Santana

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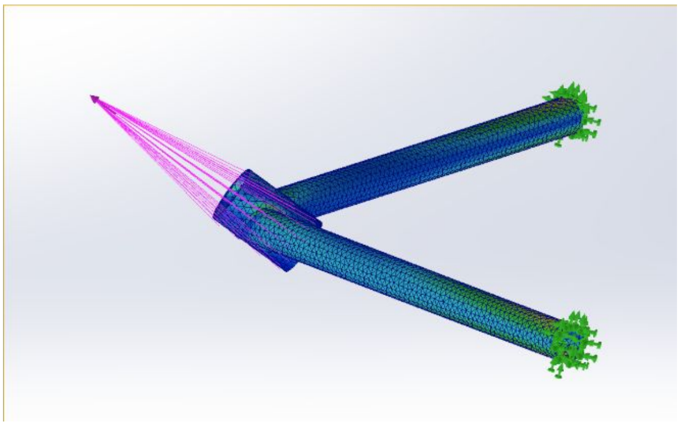
Berkeley FORMULA RACING

Berkeley Formula Racing

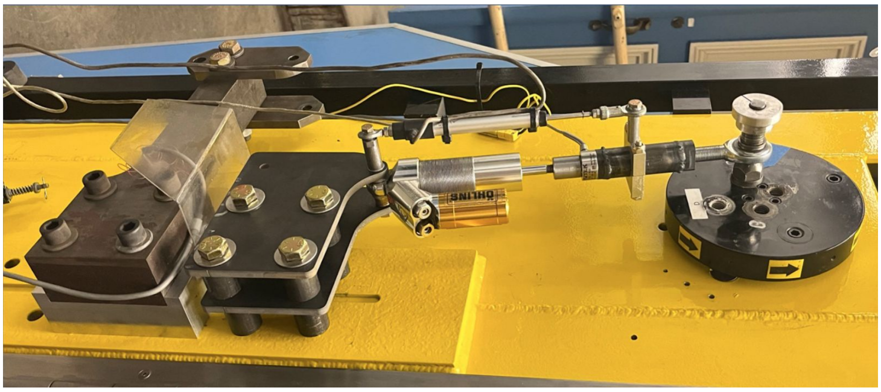
September 2025- Current
Suspension Subteam Member

Projects:

Steering Rack



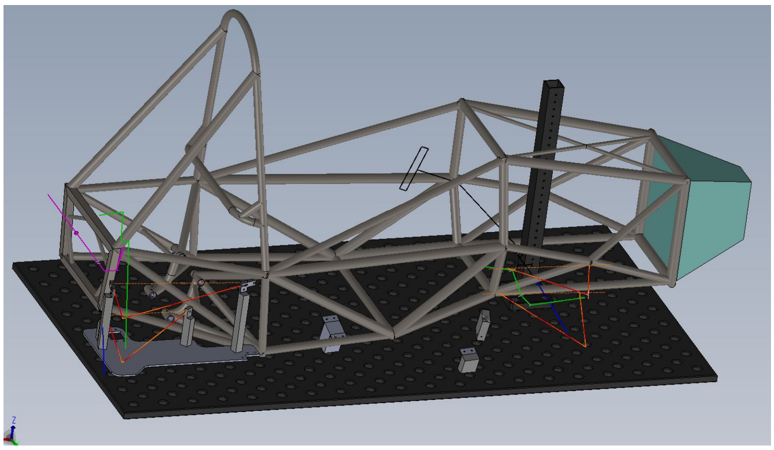
• Designed and packaged steering rack for the 2025 race car
• Modeled housing, mounts, and tie rod interfaces in SolidWorks
• Produced GD&T based drawings for critical fits
• Reduced steering play and improved serviceability



Shock Dyno

• Designed and built a suspension dyno for damper testing
• Created welding jig and sensor mounts for accurate alignment
• Integrated FUTEK load cell with Python based data workflow
• Improved repeatability and reduced setup time for testing

Welding Jigs and Fixtures



• Designed welding jigs for chassis and subassembly fabrication
• Modeled fixtures with locators, clamps, and datum references
• Validated weld access, sequence, and tolerance stack up
• Improved alignment accuracy and reduced rework

MotoStudent

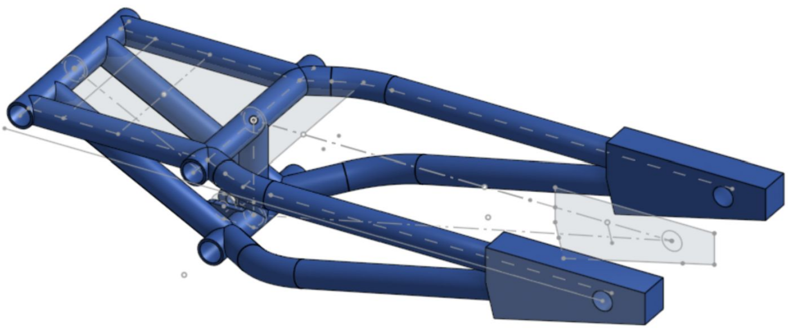
September 2024- May 2025

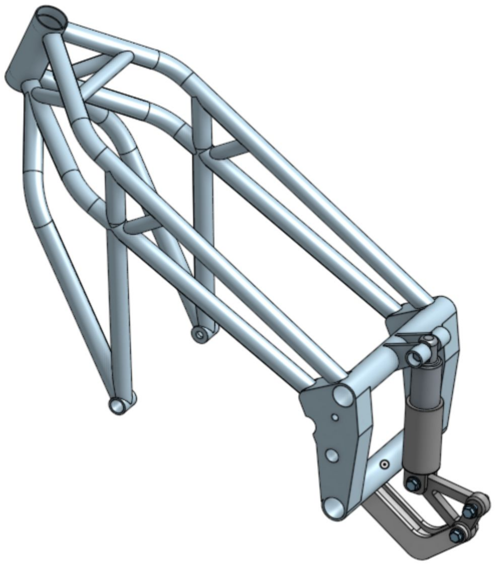
Design and Fabrication Member

Projects:

Swing Arm Design

• Designed rear swing arm geometry for race motorcycle
• Positioned pivot, axle, and shock mounts to meet chassis targets
• Checked motion, chain, tire, and brake clearances
• Prepared manufacturable design for FEA and fabrication



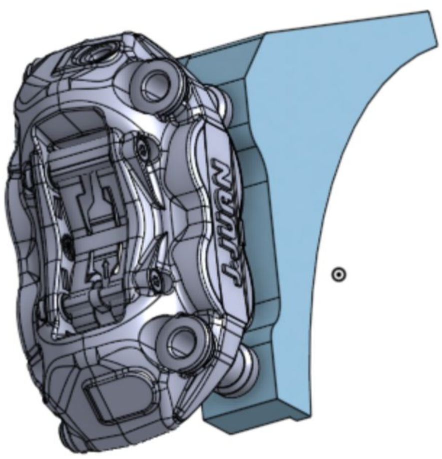


Link and Rocker Design

• Designed rear suspension link and rocker geometry for motorcycle
• Iterated leverage curves to refine motion ratio and travel
• Verified alignment and full stroke clearance in CAD
• Selected standard hardware and bearings for serviceability

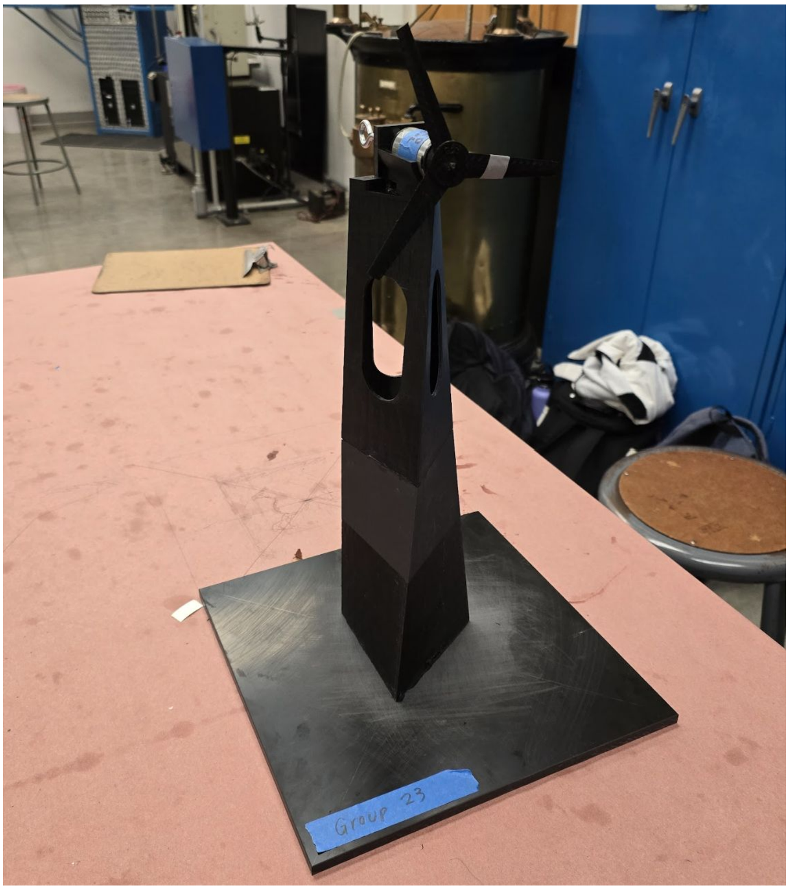
Rear Brakes

• Designed rear brake caliper mount for race motorcycle
• Modeled caliper bracket, rotor spacing, and fastener layout
• Verified rotor, caliper, and wheel hub clearances
• Simplified machining and improved pad access for service



Wind Turbine Project

September 2024- December 2024



For my introductory mechanical design course at UC Berkeley, I worked in a small team to design, 3D-print, and test a **functional wind turbine prototype**. The project focused on **aerodynamics, design for manufacturing, and system efficiency**.

I modeled the blades, hub, and housing in **SolidWorks**, optimizing the blade profile for consistent rotation under low wind speed conditions. I adjusted the hub geometry for proper balance and reduced print time by re-parameterizing support angles and wall thickness.

The finished turbine successfully generated measurable output during lab testing. This project strengthened my understanding of design iteration, printing tolerances, and performance testing with low-cost components.

Tea Kettle Project

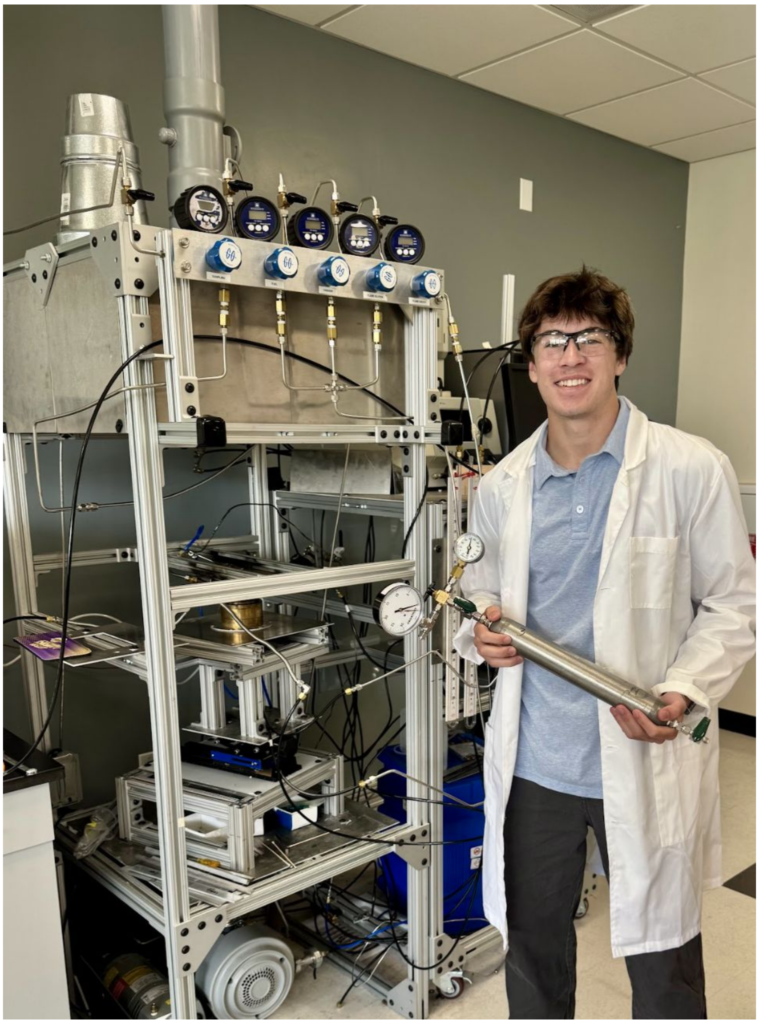
September 2025- December 2025

Designed a **screw valve attachment** for standard kettles to provide **precise, leak free flow control** for tasks such as pourover coffee. Contributed across the full design cycle, including **concept development**, **CAD iteration**, **tolerance definition**, and **assembly drawings**.

Worked with the team to **prototype and refine** the valve using laser cut plates, welded fixtures, machined housing components, and a **silicone gasket seal**. Focused on **manufacturability**, **GD&T**, and **realistic process limits**, with improvements driven by hands on testing and multiple prototype iterations. The final design balanced **cost**, **usability**, and **serviceability** and was presented at the Jacobs Showcase.

SDSU Camacho Lab

May 2025 - August 2025



At San Diego State University, I worked under Professor Joaquin Camacho supporting experimental research on flame synthesis and combustion materials. My role combined hands on lab assistance with digital design work for lab outreach.

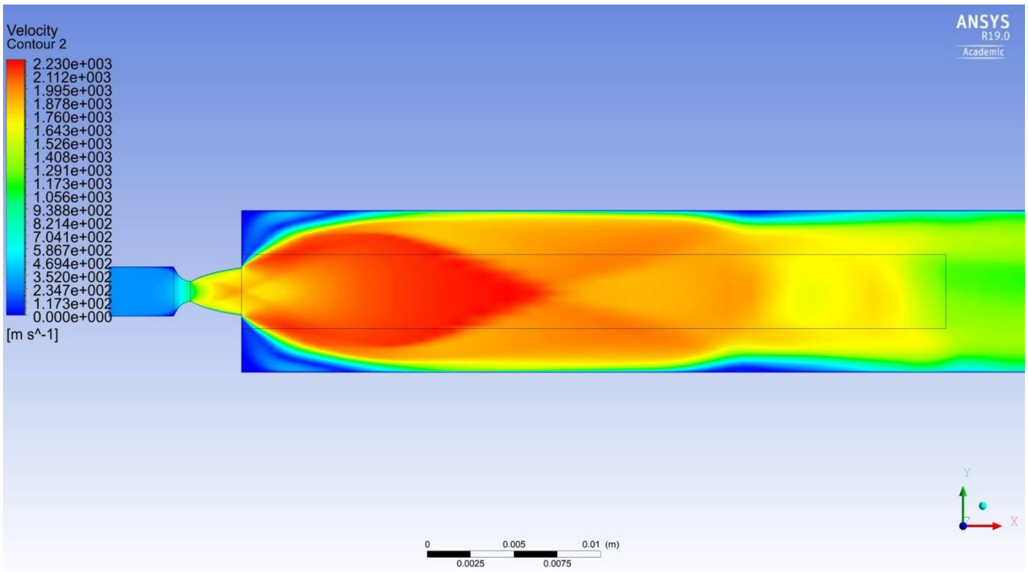
I helped set up and maintain experimental equipment for soot characterization and material synthesis, including operation of a **mass spectrometer** for species analysis, while following safety and calibration procedures. Alongside lab work, I designed and launched the group’s academic website to present publications, research areas, and team members.

This experience strengthened my lab discipline, technical communication, and ability to present research work through clear visuals for academic and public audiences.

Website: https://camacho.sdsu.edu/index.html

SDSU Dr. Adibi Lab

May 2025 - August 2025



Under **Professor Sara Adibi** at San Diego State University, I contributed to computational mechanics research focused on **aerospace materials and thermoplastic composites**. My work supported both **numerical simulation and technical proposal development** for NASA and DoD applications.

I assisted in setting up and running **molecular dynamics simulations** using **LAMMPS** to study high-temperature polymer behavior and interface strength under load. I analyzed simulation data using Python and helped generate figures for manuscripts and reports.

In addition to research tasks, I played a key role in proposal preparation, including **literature reviews, technical editing, and figure design** for NASA's MPLAN challenge and DoD SBIR Phase I submissions. This experience strengthened my ability to connect computational modeling with real engineering objectives and mission needs.

Tools: LAMMPS, Python, LaTeX, ANSYS Fluent, Microsoft Office

Will Santana

My engineering work blends **hands-on design, research, and system integration**. Whether building testing equipment for Formula SAE, designing suspension systems for MotoStudent, or contributing to computational research at SDSU, I focus on solving real mechanical challenges through collaboration and precision.

Each project on this page represents a stage in my growth as an engineer, moving from CAD design and fabrication to simulation, testing, and proposal development. I value projects that connect technical work with real-world application, and I continue to pursue opportunities that combine **design, analysis, and innovation**.

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